

Mobile Devices and Applications

UNIT 5 –UI & CONTROLS

Summary

User interface

Controls

Handlers

USER INTERFACE

User interface – UI

User interface

- visual controls
- containers

UI with the help of the designer tool

- Visually with the designer windows
- Manually by creating the XML file

UI in Java code

User interface

Easy and productive mechanism for designing the UI

Just one designed layout can start first as the layout associated with the main launcher activity

Controls can be initialized by

- by declaring them in the XML
- by declaring them in Java code

Visual components (widgets) – controls

Organizing controls (ViewGroup/Layout) – containers

User interface

View

- Used for designing the interface and handling the events
- Base class for simple visual components like widgets
- package android.view

ViewGroup

- Extends the **View** class
- Base class for complex visual controls
- package android.view

Containers

Define the visual structure of the UI

Manages controls in a:

- Linear layout (vertical/horizontal)
- Table layout
- Relative layout

Code and visual declaration

Containers

LinearLayout

- Horizontal or vertical

TableLayout

- Rows and columns
- TableRow

GridLayout (from API 14)

- Rows and columns
- Combines TableLayout with LinearLayout

RelativeLayout

- Relative to other components

ConstraintLayout

- Adds dependencies between each visual component

Containers

FrameLayout

- Visual components have stack based management

ScrollView /HorizontalScrollView

- The content can be scrolled horizontally or vertically

AbsoluteLayout (deprecated)

- Absolute positioning

Controls

android.widget.* package;

Includes

- Static text
- Editable text
- Buttons
- Lists
- Progress bars
- Etc.

Controls

TextView

- Static text

Text static

TextView

EditText

- Used as an input control
- Used for filtering input content (inputType)

Text editabil

AutoCompleteTextView

- Extends EditText class
- Binds to a list of suggestions

cl

Cluj-Napoca

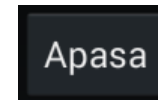
Space (from API 14)

- Used for inserting spaces between components

Controls

Button

- Standard buttons
- Extends TextView class



RadioButton

- A button used for exclusive selection
- Extends TextView class
- Regularly include in a **RadioGroup**;



CheckBox

- A button used for multiple selection



ToggleButton

- Two states button (pushed or not)



Switch (from API 14)

- Also a two states button (left-right)

Controls

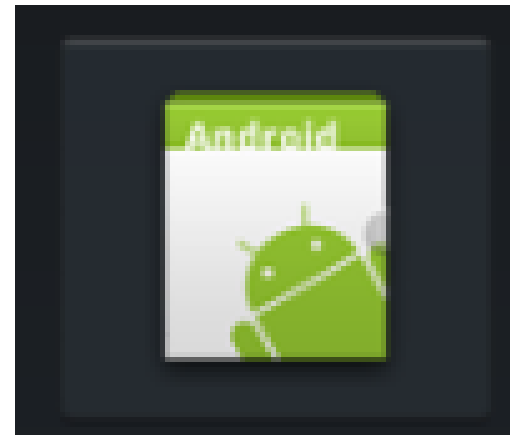
ImageView

- For displaying images



ImageButton

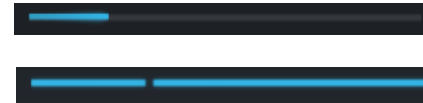
- Button functionality with images on background
- Extends an **ImageView**



Controls

ProgressBar

- Linear progress bar with upper limit
- No upper limit, cyclic



SeekBar

- Extends *ProgressBar*, used for choosing a current value



Controls

AnalogClock



DigitalClock/TextClock

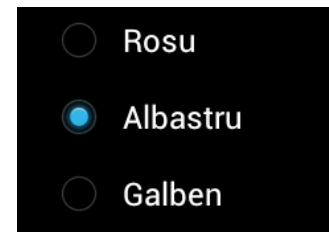
15:36

Chronometer

00:00

RadioGroup

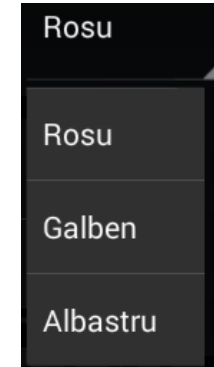
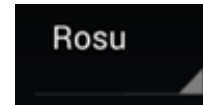
- Grouping RadioButtons



Controls

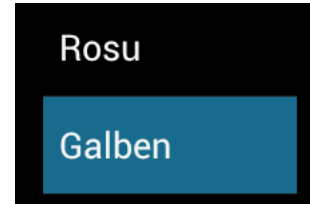
Spinner

- Control used for displaying a list of values



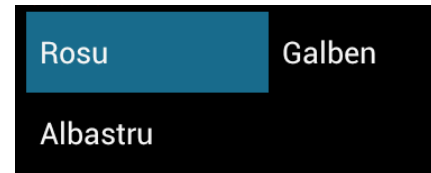
ListView

- A list of elements displayed vertically



GridView

- Displays elements using rows and columns



Complex controls

ExpandableListView

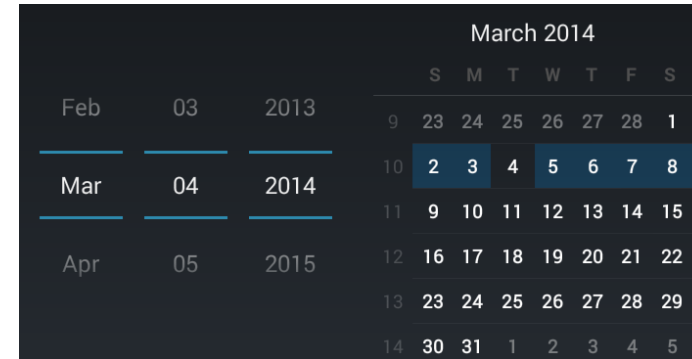
- A list with multiple elements grouped by category



Complex controls

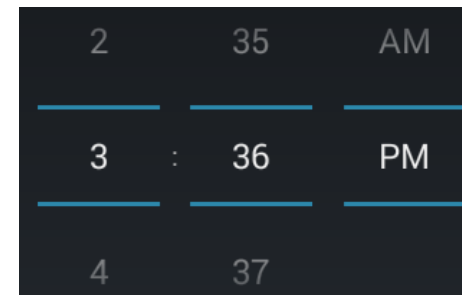
DatePicker

- Date selection
- Extends FrameLayout



TimePicker

- Time selection
- Extends also FrameLayout



Android Support Library

A library with additional classes

Used for backwards compatibility in Android

- Fragments, action bar, toolbar, etc.

Includes also some new components

- ViewPager, DrawerLayout
- RecyclerView
- Floating button
- Snackbar
- Toolbar with motion

Events handling

Listener interfaces

- Ex: **View.OnClickListener**

Methods for handling events

- Ex.: **onClick()**

Managing events

- Ex.: **setOnClickListener()**

Events handling

Interface	Event handler	Event registration
View.OnClickListener	onClick()	setOnClickListener()
View.OnKeyListener	onKey()	setOnKeyListener()
View.OnTouchListener	onTouch()	setOnTouchListener ()
View.OnLongClickListener	onLongClick()	setOnLongClickListener()

Events handling

Implementing the interface

- activity class
- special class

Implementing the interface in an anonymous class

The fast way for the onClick event:

- The onClick property of an object in the XML file
- a special method designed in the activity class

Events handling

```
public class Activitate extends Activity implements
    View.OnClickListener {

    public void onCreate(Bundle savedInstanceState) {

        //event registration
        boton.setOnClickListener(this);
        //...
    }

    //event handler
    public void onClick(View view) {
        // must test from which view the event is coming
    }
}
```

Events handling

```
public class EvenimClick implements View.OnClickListener {  
    //event handler  
    public void onClick(View view) {  
        // handling the event  
    }  
}  
  
public class Activitate extends Activity {  
    public void onCreate(Bundle savedInstanceState) {  
        //event registration  
        buton.setOnClickListener(new EvenimClick());  
        //...  
    }  
}
```

Events handling

//anonymous class

```
buton.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        // handling the event  
    }  
});
```

Events handling

XML

- android:onClick="**methodName**"

The activity class:

- public void **methodName**(View view) { ... }

The method can be used for multiple controls

- Testing the control which sends the event
 - view.getId()

Resources

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